

VOLUME 2

COMPLETE VOLUME — CHAPTERS 2.1–2.10

Product Design

How the Product Looks, Feels, and Behaves

Document Title	Volume 2 — Product Design (Combined)
Volume Reference	Volume 2
Chapters Included	2.1–2.5, 2.7, 2.9, 2.10 (8 Word chapters); 2.6 and 2.8 delivered separately as interactive HTML
Version	1.0
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Governed By	Volume 0 — Project Foundation; grounded in Volume 1 — Product Planning

Table of Contents

This volume defines how the Human Archive Collector and Admin experiences look, feel, and behave, built directly on the requirements, personas, and roles established in Volume 1. It contains the following 10 chapters. Eight are Word documents, combined into this single file below; two — Chapter 2.6 (Wireframes) and Chapter 2.8 (Design System) — are delivered as standalone interactive HTML mockups, per the project owner's own choice of format for the visual deliverables, and are referenced throughout the Word chapters rather than duplicated here.

Chapter	Title	Format
2.1	User Journey Map	Word (this file)
2.2	User Flow	Word (this file)
2.3	Information Architecture	Word (this file)
2.4	Navigation Structure	Word (this file)
2.5	Screen Inventory	Word (this file)
2.6	Wireframes	Interactive HTML — delivered separately (wireframes.html)
2.7	UI Specification	Word (this file)
2.8	Design System	Interactive HTML — delivered separately (design_system.html)
2.9	UX Guidelines	Word (this file)
2.10	Accessibility	Word (this file)

A Note on the Two HTML Chapters

- Chapter 2.6 (Wireframes) is an interactive, clickable mockup tool covering 14 of the 23 screens named in Chapter 2.5's Screen Inventory, built on the same visual language as Chapter 2.8.
- Chapter 2.8 (Design System) is the living reference for every color, type style, spacing value, and component (buttons, pills, form fields, checklist items) used throughout Chapters 2.6 and 2.7, in both light and dark themes.
- Both were built as interactive HTML rather than static Word documents by explicit choice, since a design system and a wireframe set are reference tools meant to be opened, clicked through, and revisited — not printed.

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CHAPTER 2.1

User Journey

Walking a Day in Riya's and Aman's Shoes

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Chapter Reference	2.1
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Based On	Chapter 1.5 — User Personas ("Riya" the Collector, "Aman" the Admin)

1. Purpose

This chapter walks through the end-to-end experience for both current, MVP-scope roles — the Collector and the Admin — stage by stage, surfacing what each person is trying to do, what they're thinking and feeling, and where the design needs to actively help rather than just get out of the way. It is the emotional/experiential companion to Chapter 2.2 (User Flow), which covers the same ground mechanically.

2. Collector Journey — "Riya"

Riya is in the field, often alone, often with unreliable connectivity (Chapter 1.5). Her journey runs once per work session, several times a day.

2.1 Arriving at a New Task

Stage Goal: Understand exactly what to record before starting.

What They're Doing: Opens the app, checks the Dashboard, opens her Project, opens a Task.

What They're Thinking: "Do I know exactly what's expected here? Have I done this Task before?"

How They Feel: Neutral to slightly anxious — she doesn't want to redo work because she misread instructions.

Pain Point If Unaddressed: If instructions are vague or examples are missing, she guesses — and may record the wrong thing entirely.

Design Opportunity: Task Detail must front-load instructions and reference examples before any recording action is reachable (Chapter 1.3, FR-PT-05).

2.2 Getting Ready to Record

Stage Goal: Confirm the device and environment are actually ready.

What They're Doing: Proceeds to the Pre-Recording Checklist; the app checks permissions, storage, battery, network.

What They're Thinking: "Is my phone going to make it through this recording? Do I have enough space?"

How They Feel: Slight tension — she has been burned before by a recording that failed partway through.

Pain Point If Unaddressed: A checklist that fails silently, or blocks her without saying why, reads as the app being broken rather than helpful.

Design Opportunity: Every checklist failure names the exact problem and the exact fix (Chapter 1.3, FR-CHK-05) — never a bare "cannot record" message.

2.3 Recording

Stage Goal: Capture the required footage correctly, without fiddling with the app mid-task.

What They're Doing: Full-screen preview, taps Record, performs the task, taps Stop.

What They're Thinking: "Is it actually recording? How long has it been?"

How They Feel: Focused — this is the one moment she needs the UI to disappear and just confirm it's working.

Pain Point If Unaddressed: Ambiguous recording state (is it on? paused? crashed?) erodes trust in the entire session.

Design Opportunity: A persistent, unmistakable recording indicator and timer (Chapter 1.3, FR-REC-04); nothing else competes for attention on this screen.

2.4 Letting Go of the Upload

Stage Goal: Trust that stopping the recording did not put the footage at risk, and get back to work.

What They're Doing: Taps Stop; the app chunks locally and begins uploading; she moves to her next Task.

What They're Thinking: "Is my footage safe now? Can I already start the next one?"

How They Feel: Relief, if the app confirms safety immediately; renewed anxiety if it goes quiet.

Pain Point If Unaddressed: If the app appears to do nothing after Stop, she doesn't know whether to trust it, retry, or wait.

Design Opportunity: Immediate, visible confirmation that the recording was safely persisted (Chapter 1.3, FR-REC-06) before she's ever asked to think about upload at all.

2.5 Working Through Bad Connectivity

Stage Goal: Keep recording even when the network can't keep up.

What They're Doing: Starts another recording while the previous one is still uploading, possibly offline.

What They're Thinking: "Am I going to lose this work because the last one hasn't finished?"

How They Feel: Wary — this is the moment the product either proves itself or loses her confidence.

Pain Point If Unaddressed: Being blocked from recording because of someone else's (the upload engine's) problem feels arbitrary and punishing.

Design Opportunity: Recording is never gated on upload state (Chapter 1.9, BR-10); queued/offline chunks are visibly queued, not silently stuck.

2.6 Confirming the Work Is Done

Stage Goal: Know, without doubt, that a Task is genuinely finished.

What They're Doing: Checks session status; sees the session marked Complete; returns to the Task List.

What They're Thinking: "Is that it? Am I done, or is something still happening in the background?"

How They Feel: Satisfaction, if Complete is unambiguous; lingering doubt if the status is unclear.

Pain Point If Unaddressed: A vague or optimistic status (marking things 'done' before upload is actually confirmed) destroys trust the first time it's wrong.

Design Opportunity: "Complete" is reserved exclusively for fully uploaded and metadata-verified sessions (Chapter 1.9, BR-12, BR-21) — never shown early.

3. Admin Journey — "Aman"

Aman structures the work before any Collector ever opens the app, and checks in on progress throughout the engagement, rather than working continuously inside it.

3.1 Standing Up a New Project

Stage Goal: Get a new piece of work organized and ready to assign quickly.

What They're Doing: Creates a Project, adds Tasks with instructions and examples.

What They're Thinking: "Will Collectors understand this without me having to explain it in person?"

How They Feel: Focused, slightly rushed — this often happens under time pressure to get collection started.

Pain Point If Unaddressed: If task creation is slow or unclear, Collectors get assigned incomplete or ambiguous work.

Design Opportunity: Task creation mirrors exactly what a Collector will see (Chapter 1.3, FR-ADM-02) so Aman can preview the Collector's experience while authoring it.

3.2 Assigning the Right People

Stage Goal: Get the work to the right Collectors without back-and-forth.

What They're Doing: Selects Collectors and assigns them to the Project/Task.

What They're Thinking: "Does this person have the right device/skills for this? Will they see it right away?"

How They Feel: Confident, as long as assignment is immediate and visible.

Pain Point If Unaddressed: An assignment that doesn't show up promptly on the Collector's side creates confusion and duplicate outreach (calls, messages) to confirm it worked.

Design Opportunity: Assignment is immediately reflected and confirmable from the Admin side (Chapter 1.3, FR-ADM-08).

3.3 Checking In on Progress

Stage Goal: Know how collection is going without chasing anyone.

What They're Doing: Opens Session/chunk status views for Projects they manage.

What They're Thinking: "Are we on track? Is anyone stuck? Is this actually the footage I asked for?"

How They Feel: Calm if status is current and trustworthy; frustrated if it's stale or ambiguous.

Pain Point If Unaddressed: Status that doesn't reflect what's genuinely uploaded and verified leads to false confidence — and late discovery of missing data.

Design Opportunity: Status shown to Aman is the same verified truth shown to the Collector — never an optimistic preview (Chapter 1.3, FR-ADM-05).

3.4 Verifying the Work Is Real

Stage Goal: Trust that footage marked complete is genuinely usable.

What They're Doing: Opens metadata for a chunk/session — timestamps, device, GPS, checksum status.

What They're Thinking: "Is this actually what it claims to be? Right project, right time, right person?"

How They Feel: Reassured once metadata is visible and complete; suspicious if it's partial or missing.

Pain Point If Unaddressed: Without verifiable metadata, Aman has no way to catch a problem before it reaches downstream dataset consumers.

Design Opportunity: Full metadata is viewable/exportable per chunk and session (Chapter 1.3, FR-ADM-06; Chapter 1.9, BR-21–23).

4. Cross-Cutting Observation

Both journeys converge on the same underlying need: certainty. Riya needs certainty that her recording is safe the instant she stops it. Aman needs certainty that what's marked complete is actually, verifiably complete. Every design decision in this volume is judged against that standard — does it make the true state of the system obvious, or does it ask someone to just trust it?

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CHAPTER 2.2

User Flow

Every Path, Including the Branches

Document Title	User Flow
Chapter Reference	2.2
Version	1.0
Status	Draft — Pending Approval

Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Expands	Chapter 1.1, Section 4 (End-to-End Product Flow) with decision branches

1. Purpose

Chapter 1.1 confirmed the linear, happy-path flow. This chapter expands every step that has a decision point into its success and failure/alternate paths, so Volume 2.6 (Wireframes) and later engineering volumes have no ambiguity about what happens when something doesn't go perfectly — which, in the field, is often.

2. Collector Flow

Step	Screen / Action	On Success →	On Failure / Branch →
1	Install App	Onboarding begins	—
2	Onboarding / Permission Priming — requests Camera, Microphone, Location, Notifications, Files	All granted → Login	Any required permission denied → Permission Blocked screen (explainer + settings deep link); cannot proceed to recording until resolved
3	Login (credentials or SSO)	Session established → Home Dashboard	Invalid credentials → inline error, retry; SSO failure → provider's error shown, retry
4	Home Dashboard	Selects a Project → Projects List	No Projects assigned → empty state with guidance to contact an Admin
5	Projects List → Project Detail → Task List	Selects a Task → Task Detail	No Tasks in Project → empty state
6	Task Detail (instructions, examples, requirements)	Proceeds → Pre-Recording Checklist	—
7	Pre-Recording Checklist (permissions, storage, battery, network)	All checks pass → Recording Screen	Any check fails → Checklist Failed screen with the specific remedy; re-run checklist after fixing
8	Recording Screen — Record tapped	Recording in progress, timer running	Camera/mic error mid-start → error shown, return to Task Detail without a partial session
9	Stop Recording	Raw footage safely persisted → Local Processing	App killed before Stop → footage recorded up to last safe write point is preserved (Chapter 1.9, BR-13); resumes into Local Processing on next launch
10	Local Processing — Automatic Chunking + Metadata Generation	Chunks + metadata persisted locally → Upload begins	—
11	Upload (automatic default, or manual if configured)	Network available → chunks upload in background	Offline → chunks queue locally; Collector free to start a new recording (Chapter 1.9, BR-10)
12	Upload Progress / Sync Status	All chunks confirmed + metadata verified → Session Marked Complete	A chunk fails after retries → shown as Failed with a manual Retry action (Chapter 1.8, UC-06); session stays open, not Complete
13	Session Marked Complete	Return to Task List / Dashboard	—

3. Admin Flow (New — MVP)

Step	Screen / Action	On Success →	On Failure / Branch →
1	Login (credentials or SSO)	Session established → role routed to Admin Dashboard	Same failure branches as Collector Login (Section 2, Step 3)

Step	Screen / Action	On Success →	On Failure / Branch →
2	Admin Dashboard	Selects "New Project" or an existing Project	No Projects yet → empty state prompting Project creation
3	Create/Edit Project	Saved → Project Detail (Admin)	Required fields missing → inline validation, blocks save
4	Project Detail (Admin) — Task List	Selects "New Task" or an existing Task	—
5	Create/Edit Task (instructions, examples, requirements)	Saved → Task appears in Project Detail, not yet visible to any Collector	Required fields missing → inline validation, blocks save
6	Assign Collectors to the Task	Collector(s) selected and confirmed → Task becomes visible in those Collectors' Task Lists (Chapter 1.3, FR-ADM-08)	Selected Collector has no account / is deactivated → blocked with explanation (Chapter 1.8, UC-07)
7	Session/Chunk Status view (per Task)	Views live status across all assigned Collectors	No sessions yet → empty state, "no activity yet"
8	Metadata Detail / Export (per chunk or session)	Views or exports the full metadata record	Metadata not yet confirmed by backend → shown as "pending verification," not yet exportable

4. Notes on Branch Handling

- No failure branch ever silently discards user input or recorded footage — every failure path either blocks with an explanation or preserves state for retry (Chapter 1.9, Business Rules).
- Any branch that can recur (checklist failure, upload failure) loops back to the same step rather than forcing a restart of the whole flow.
- Both flows share the same Login step and its failure branches — see Section 2, Step 3 — rather than duplicating them per role.

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CHAPTER 2.3

Information Architecture

How Content and Screens Are Organized

Document Title	Information Architecture
Chapter Reference	2.3
Version	1.0
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)

Date	05 August 2026
Companion Document	Chapter 2.5 — Screen Inventory (flat list of every screen named here)

1. Purpose

This chapter defines the hierarchy of content and screens for both roles — what sits inside what, and what a person can reach from where. Chapter 2.4 (Navigation Structure) then defines how a person physically moves through this hierarchy (tabs, stacks, modals).

2. Collector Information Architecture

- Collector Root
 - Home Dashboard
 - Active Projects summary
 - In-progress Sessions summary
 - Total recorded time, sync status
 - Projects
 - Project
 - Tasks
 - Task
- Instructions / Examples / Requirements
 - Pre-Recording Checklist
 - Permissions / Storage / Battery / Network checks
 - Recording Screen
 - Local Processing (chunking + metadata)
 - Sessions
 - Session
 - Chunks & upload status
 - Notes / tags (Phase 2)
 - Settings
 - Account / logout
 - Upload mode (automatic / manual)
 - Support / help contact (Phase 2)
 - Notifications (Phase 2)

3. Admin Information Architecture

- Admin Root

- Admin Dashboard
 - Managed Projects summary
 - Collector activity summary
- Projects
 - Project
- Create / Edit Project
 - Tasks
 - Task
- Create / Edit Task
 - Collector assignment
- Assign / reassign / remove Collector
 - Sessions & Metadata (per Project)
 - Session / chunk status
 - Metadata detail / export
 - Advanced Ops (Phase 3 — not MVP)
 - Fleet-wide monitoring
 - QA / rework flagging
 - Collector performance analytics
 - Remote capture configuration

4. Shared Nodes

Both role trees share a single entry point rather than maintaining two separate apps or login systems:

- Login (credentials or SSO) — shared screen, role-agnostic.
- Role Router — the only node that differs post-login; routes to Collector Root or Admin Root based on the authenticated user's role (Chapter 1.6).
- Account/session management (token refresh, logout) — shared logic, role-specific presentation.

5. Depth & Reachability Rules

- No screen in the Collector tree is reachable from the Admin tree, or vice versa, except through logout → re-login as a different role.

- A Collector can never reach a Project/Task they are not assigned to — the Projects node under Collector Root only ever contains assigned items (Chapter 1.9, BR-14, BR-19).
- An Admin can never reach Projects outside their own scope — the Projects node under Admin Root is scoped to Projects they manage (Chapter 1.9, BR-20).
- The Recording Screen is reachable only through the Pre-Recording Checklist — there is no direct path to it from the Dashboard or Task List (Chapter 1.9, BR-04).

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CHAPTER 2.4

Navigation Structure

How People Physically Move Through the App

Document Title	Navigation Structure
Chapter Reference	2.4
Version	1.0
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Companion Document	Chapter 2.3 — Information Architecture

1. Purpose

Chapter 2.3 defined what's inside what. This chapter defines how a person moves between those things — which parts are persistent chrome (tabs), which are drill-down stacks, and which interrupt the flow as modals. Volume 6 (Mobile App Architecture) will implement this as a Flutter navigator/router structure.

2. Navigation Model — Collector

Root pattern: Bottom tab bar, persistent across the Collector experience.

- Tab 1 — Dashboard (Home)
- Tab 2 — Projects
- Tab 3 — Record (shortcut — jumps into the most relevant in-progress Task's checklist, or prompts Task selection if none is obviously in progress)
- Tab 4 — Sessions (upload/sync status across all sessions)
- Tab 5 — Settings

Stack navigation (drill-down, back-swipe/back-button enabled): Projects → Project Detail → Task List → Task Detail. Sessions → Session Detail → Chunk Detail.

Modal presentation (interrupts the stack, must be resolved or dismissed): Pre-Recording Checklist (presented full-screen modal from Task Detail); Checklist Failed remedy screen (modal-within-modal, or replaces checklist content in place); Permission Blocked screen (full-screen modal, shown at first launch or whenever a required permission is later revoked).

Full-screen, chrome-free (no tab bar, no back button during capture): Recording Screen — deliberately removes all navigation chrome so nothing can be tapped accidentally mid-recording; only Stop is reachable.

3. Navigation Model — Admin

Root pattern: Bottom tab bar (mobile) or left sidebar (if a web surface is chosen — Chapter 1.1, Section 8 open question), persistent across the Admin experience.

- Tab/Section 1 — Dashboard (Home)
- Tab/Section 2 — Projects (create/edit Projects & Tasks, assign Collectors)
- Tab/Section 3 — Sessions & Metadata (status + metadata across managed Projects)
- Tab/Section 4 — Settings/Account

Stack navigation: Projects → Project Detail → Task List → Task Detail → Assign Collectors.

Modal presentation: Create/Edit Project; Create/Edit Task; Assign Collectors picker; Metadata Detail/Export view (presented as an overlay on top of Session/Chunk Status so Admin doesn't lose their place).

4. Role Switching & Entry

- Login is a single, shared screen for both roles — there is no separate "Admin login" surface.
- Immediately after authentication, the Role Router (Chapter 2.3, Section 4) sends the user to the Collector root or Admin root — this happens once, silently, and is not user-visible as a separate step.
- A user holding both roles (rare, but not excluded architecturally) would need an explicit role switch affordance in Settings — out of scope for MVP; flagged as an open question for Volume 6 if it becomes a real requirement.

5. Navigation Summary Table

Pattern	Used For	Notes
Bottom tab bar	Collector & Admin root-level sections	Persistent, always reachable except during active Recording
Stack (drill-down)	Projects → Tasks → Detail; Sessions → Chunks	Standard back-navigation; state preserved on back
Modal / full-screen overlay	Checklist, Permission Blocked, Create/Edit forms, Metadata Detail	Must be explicitly resolved or dismissed; not swappable mid-task
Chrome-free full-screen	Recording Screen only	No tabs, no back button; only Stop is reachable, by design (Chapter 1.9, BR-01–04)

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CHAPTER 2.5

Screen Inventory

Every Screen the Product Needs, Named and Scoped

Document Title	Screen Inventory
Chapter Reference	2.5
Version	1.0
Status	Draft — Pending Approval
Owner	Faisal — Human Archive

Prepared With	Claude (Cowork)
Date	05 August 2026
Purpose	The master, authoritative list of screens — Chapter 2.6 (Wireframes) and Chapter 2.7 (UI Specification) both reference these IDs directly.

1. Shared Screens (SH)

ID	Screen Name	Role	Purpose / Key Content
SH-01	Splash / App Launch	Shared	Brief branded loading state while session/token state resolves.
SH-02	Login	Shared	Email/password fields, SSO entry point, error states (Chapter 2.2, Section 2 Step 3).
SH-03	Role Router	Shared (system)	Not user-facing — silently routes to Collector Root or Admin Root post-login.

2. Collector Screens (C)

ID	Screen Name	Role	Purpose / Key Content
C-01	Onboarding / Permission Priming	Collector	Carousel explaining why Camera, Microphone, Location, Notifications, Files are needed, before the OS prompts appear.
C-02	Permission Blocked	Collector	Blocking explainer + settings deep link when a required permission is denied (Chapter 1.3, FR-ONB-02).
C-03	Home Dashboard	Collector	Active Projects, in-progress Sessions, total recorded time, sync status, pending/uploading/completed chunk counts.
C-04	Projects List	Collector	Every Project assigned to this Collector.
C-05	Project Detail	Collector	Task List within the selected Project.
C-06	Task Detail	Collector	Instructions, reference examples, requirements; entry point to the Checklist.
C-07	Pre-Recording Checklist	Collector	Live permission/storage/battery/network checks (Chapter 1.3, FR-CHK).
C-08	Checklist Failed	Collector	Names the specific failed check and the fix; re-run action.
C-09	Recording Screen	Collector	Full-screen wide-angle preview, Record/Stop, elapsed timer, recording indicator. Chrome-free (Chapter 2.4).
C-10	Local Processing	Collector	Brief transient state while chunking + metadata generation run after Stop.
C-11	Upload / Sync Status	Collector	Per-session, per-chunk status: queued / uploading / failed / complete.
C-12	Session Complete	Collector	Confirms every chunk + metadata verified; returns to Task List/Dashboard.
C-13	Session Notes / Tag	Collector (Phase 2)	Optional free-text notes/tags attached to a session (Chapter 1.3, FR-SEC-01).
C-14	Settings	Collector	Account/logout, upload mode (automatic/manual), support contact (Phase 2).
C-15	Notifications	Collector (Phase 2)	List of push notifications — upload failures, new assignments.

3. Admin Screens (A)

ID	Screen Name	Role	Purpose / Key Content
A-01	Admin Dashboard	Admin	Managed Projects summary, Collector activity summary, outstanding Task counts.
A-02	Projects List (Admin)	Admin	Every Project this Admin manages.
A-03	Project Detail (Admin)	Admin	Task list within the Project; entry points to create/edit and to Collector assignment.
A-04	Create / Edit Project	Admin	Name, description, and Project-level settings.
A-05	Create / Edit Task	Admin	Instructions, reference examples, requirements — mirrors what the Collector will see (Chapter 1.3, FR-ADM-02).
A-06	Assign Collectors	Admin	Select/reassign/remove Collectors for a Project or Task (Chapter 1.8, UC-07).

ID	Screen Name	Role	Purpose / Key Content
A-07	Session / Chunk Status (Admin)	Admin	Live status across all Collectors for a given Project/Task (Chapter 1.3, FR-ADM-05).
A-08	Metadata Detail / Export	Admin	Full metadata record for a chunk or session; export action (Chapter 1.3, FR-ADM-06).

4. Deferred Screens (Phase 3 — Not MVP, Named for Forward Compatibility)

ID	Screen Name	Role	Purpose / Key Content
A-09	Fleet Monitoring Dashboard	Super Admin (Future)	Live, fleet-wide view across every Project/Collector (Chapter 1.11, Phase 3).
A-10	QA / Rework Queue	Admin (Future)	Flag sessions/chunks for rework; track resolution.
A-11	Collector Performance Analytics	Admin (Future)	Throughput and reliability analytics per Collector.
A-12	Remote Capture Configuration	Super Admin (Future)	Chunk duration/resolution/bitrate feature flags (Chapter 1.11, Phase 3).

5. Screen Count Summary

23 screens total for the MVP-relevant scope (3 shared + 15 Collector + 8 Admin, one of which — A-08 — doubles as the metadata surface referenced throughout Volume 1), plus 4 named-but-deferred Phase 3 screens kept here so the IA (Chapter 2.3) and future Volume 6 navigation graph don't need restructuring when Phase 3 begins.

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CHAPTER 2.7

UI Specification

Screen-by-Screen Component Behavior, Built on the Design System

Document Title	UI Specification
Chapter Reference	2.7
Version	1.0
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Companion Documents	Chapter 2.5 (Screen Inventory), Chapter 2.6 (Wireframes), Chapter 2.8 (Design System)

1. Purpose & How to Read This Chapter

Chapter 2.5 named every screen. Chapter 2.6 showed its structural layout. Chapter 2.8 defined the reusable visual components (buttons, pills, checklist items, form fields). This chapter is where those three meet: for each screen that carries meaningful interaction logic, it specifies exactly which Design System component appears where, and what rule governs its content and behavior. Volume 6 (Mobile App Architecture) and the Flutter widget tree implement directly from these tables.

Not every one of the 23 screens in the inventory is specified individually here — screens that are pure list/detail restatements of an already-specified pattern (e.g. A-02 Projects List (Admin) follows the same rules as C-04 Projects List) are covered by cross-reference rather than a duplicate table. The screens specified in full are the ones with distinct interaction logic: onboarding, the pre-recording checklist and its failure state, the chrome-free recording screen, upload/sync status, session completion, and the two Admin screens with the most complex state — Assign Collectors and Metadata Detail.

2. Collector — Onboarding & Checklist

C-01 — Onboarding / Permission Priming

Role: Collector

Purpose: Explain, in plain language, why each OS permission will be requested — before the OS prompt itself appears — so the Collector doesn't reflexively deny it.

Layout: Full-screen carousel, one permission per card, primary button advances; last card's primary button triggers the first real OS permission prompt.

Element	Component (Ch. 2.8)	Behavior / Content Rule
Card body	Card (Design System §5, card container)	One sentence stating the permission and the concrete reason it's needed (e.g. "Camera & Microphone — to record your walkthroughs").
Progress indicator	Dot row (not in Design System v1.0 — flagged as a net-new component for the Ch. 2.8 v1.1 pass)	Shows position in the carousel; never a percentage, since the count is fixed and small.
Primary button	btn-primary	Label changes per card: "Next" on cards 1–4, "Get Started" on the final card.

C-07 — Pre-Recording Checklist

Role: Collector

Purpose: Run FR-CHK-01–04 (camera/mic permission, storage, battery, network) live and show pass/fail per check before Start Recording is enabled.

Layout: Vertical list of checklist items inside a card; primary action fixed to the bottom of the screen.

States: All-pass (shown) → Start Recording enabled. Any-fail → button disabled, screen transitions to C-08.

Element	Component (Ch. 2.8)	Behavior / Content Rule
Each check row	checklist-item + check-icon (check-pass)	Icon + short label + current value (e.g. "Battery: 82%"), never a bare icon (Chapter 2.10, color-not-alone rule).
Start Recording	btn-primary, full-width	Disabled (btn-disabled) until every row shows check-pass; re-evaluates live if a value changes (e.g. battery drains) while the screen is open.

C-08 — Checklist Failed

Role: Collector

Purpose: Implement FR-CHK-05 — name the specific failed check and its remedy, never a generic error.

Layout: Same list as C-07, with the failing row expanded into an error state; a remedy card beneath it.

Element	Component (Ch. 2.8)	Behavior / Content Rule
Failing row	checklist-item + check-icon (check-fail)	Icon + label + the value that caused the failure (e.g. "Battery: 14%").
Remedy card	field.error styling (critical-colored left border, per Design System §7 error state)	One sentence, plain-language fix (e.g. "Charge to at least 20% before recording.") — never just "Error" or an error code.
Re-run action	btn-secondary	Re-runs the specific failed check only, not the full checklist, so a fixed battery doesn't force re-granting an already-granted permission.

3. Collector — Recording & Upload

C-09 — Recording Screen

Role: Collector

Purpose: The single-purpose capture surface. Deliberately minimal so nothing is tappable by accident mid-recording (Chapter 1.9, BR-01–04; Chapter 2.4, Section 2).

Layout: Full-bleed camera preview, no tab bar, no back button, no status bar chrome beyond the OS clock.

Element	Component (Ch. 2.8)	Behavior / Content Rule
Recording indicator	Net-new component — pulsing dot + elapsed timer (red, per Design System critical hue, but denoting "live" not "error" in this one context — labelled "REC" to disambiguate)	Always visible while recording; timer format mm:ss up to 9:59, then hh:mm:ss.
Stop control	btn-destructive equivalent, circular, large touch target ($\geq 64\text{px}$, exceeding the 44px minimum in Chapter 2.10)	The only interactive element on screen; ends the active chunk and triggers local chunking + metadata generation (FR-META-08).

C-11 — Upload / Sync Status

Role: Collector

Purpose: Show every chunk's status per session, mapped 1:1 to the FR-SES status values, using the fixed status palette from Chapter 2.8, Section 2.

Layout: Vertical list of chunk rows grouped under the session name; a failed row expands in place to offer retry.

Element	Component (Ch. 2.8)	Behavior / Content Rule
Chunk row (queued)	pill-queued	Warning-hue pill; label always reads "Queued", never left to color alone.
Chunk row (uploading)	pill-uploading + progress-track/progress-fill	Accent-hue pill with live percentage; progress bar beneath the row, not inside the pill.
Chunk row (failed)	pill-failed, row gets a critical-colored border	Expands to reveal a btn-secondary "Retry Chunk" action; failure never auto-retries silently (Volume 5 will define the retry/backoff policy this UI triggers).
Chunk row (complete)	pill-complete	Good-hue pill; row is otherwise inert once complete — no action needed.

C-12 — Session Complete

Role: Collector

Purpose: Give an unambiguous, satisfying confirmation that meets FR-SES-02 and BR-21 — every chunk and its metadata verified, not just uploaded.

Layout: Centered confirmation card — icon, headline, one supporting sentence, single primary action.

Element	Component (Ch. 2.8)	Behavior / Content Rule
Confirmation icon	Net-new — good-hue circular checkmark, larger than a checklist-item icon	Appears only once every chunk has cleared checksum verification (FR-META-12), never on partial completion.
Supporting text	Body text style (Design System §3)	States the chunk count explicitly (e.g. "All 6 chunks and their metadata are confirmed uploaded.") rather than a generic "Done".
Primary action	btn-primary	Returns to Task List (C-06), not the Dashboard, so the Collector's place in their assigned work is preserved.

4. Admin — Assignment & Metadata

A-06 — Assign Collectors

Role: Admin

Purpose: Implement FR-ADM-03/04 (US-29, US-30) — select, add, or remove Collectors for a Project or Task.

Layout: Scoped by Project/Task name shown at the top; vertical list of Collector rows with a checkbox each; fixed bottom confirm action.

Element	Component (Ch. 2.8)	Behavior / Content Rule
Collector row	Net-new — avatar (initial, brand-colored circle) + name + checkbox	Checkbox reflects current assignment state on load; unchecked-and-saved triggers FR-ADM-04 removal, not a silent no-op.

Element	Component (Ch. 2.8)	Behavior / Content Rule
Confirm action	btn-primary, full-width	Disabled if no change was made from the loaded state, to avoid a no-op write and a false "saved" confirmation.

A-08 — Metadata Detail / Export

Role: Admin

Purpose: Implement FR-ADM-06 — the full metadata record for a chunk or session, human-readable, per the field groups defined in Chapter 1.3, Section 6 (FR-META-01–07).

Layout: Label/value row list inside a card, grouped in the same order as FR-META-01–07 (Identity → Timing → Technical → Device → Conditions → Integrity → Collector-authored); export action fixed at the bottom.

States: Every row is read-only except the optional Collector-authored notes/tags field, per BR-22 (system-generated fields are immutable).

Element	Component (Ch. 2.8)	Behavior / Content Rule
Checksum row	field row + pill-complete ("Verified") or pill-failed ("Mismatch")	Never shown as a raw hash alone — always paired with a verified/mismatch status label so an Admin can act on it without decoding the value.
Export action	btn-secondary	Produces the export named in FR-ADM-06; exact file format is a Volume 5 (Backend) decision, not a UI one.

5. Cross-Reference — Screens Specified by Pattern, Not Duplicated

The following screens reuse a pattern already specified above or are standard list/detail/form screens with no interaction logic beyond the Design System component library; they are intentionally not given their own table to avoid restating identical rules:

- C-03 Home Dashboard, C-04 Projects List, C-05 Project Detail, C-06 Task Detail — standard card-list and label/value layouts using only components already defined in Chapter 2.8.
- A-01 Admin Dashboard, A-02 Projects List (Admin), A-03 Project Detail (Admin) — same pattern as their Collector counterparts, with Admin-only action buttons (" + New Project", " + New Task") added per FR-ADM-01/02.
- A-04 Create/Edit Project, A-05 Create/Edit Task — standard forms using the field component (Design System §7), including its built-in error state.
- SH-01 Splash, SH-02 Login, SH-03 Role Router — SH-02 uses the field component plus a single primary button and one secondary SSO entry point; SH-01 and SH-03 carry no persistent UI beyond a loading state.

6. Open Items Carried to Volume 6

- Two components used above (progress dot row for onboarding, and the recording-indicator/stop-control pairing) are net-new beyond Chapter 2.8 v1.0 — they should be folded back into the Design System in its next revision rather than treated as one-offs.
- Exact retry/backoff timing behind the C-11 "Retry Chunk" action is a Volume 5 (Backend Architecture) decision; this chapter specifies only the UI trigger, not the underlying policy.
- Export file format behind A-08's "Export Metadata" action is likewise a Volume 5 decision.

VOLUME 2 — PRODUCT DESIGN

CHAPTER 2.9

UX Guidelines

How the App Should Feel in the Moments That Matter Most

Document Title	UX Guidelines
Chapter Reference	2.9
Version	1.0
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Companion Documents	Chapter 2.7 (UI Specification), Chapter 2.8 (Design System)

1. Purpose

Chapter 2.7 specified what each component does. This chapter specifies how the app should behave and communicate in the handful of moments most likely to make or break trust: a failed checklist, a dropped connection mid-upload, an Admin removing a Collector's access, and the long stretch of nothing happening while a chunk uploads in the background. These are guidelines, not new requirements — every rule here traces back to a Functional Requirement, Business Rule, or Non-Functional Requirement already agreed in Volume 1.

2. Core Principles

1. Never fail silently. Every failure state (checklist failure, upload failure, permission denial) must name the specific cause and the specific fix, per FR-CHK-05. A generic "Something went wrong" is treated as a defect, not an acceptable fallback.

2. The Collector should never have to wonder if it worked. Recording start/stop, chunk creation, and session completion each get an explicit, unambiguous confirmation (Chapter 2.7, C-12) — the app never leaves the Collector guessing whether an action registered.

3. Background work stays visible, never invisible. Uploading is designed to happen while the Collector moves on to record something else (per the original product brief) — so upload status (C-11) must be reachable from the Dashboard in at most one tap at all times, not buried in Settings.

4. Admin actions that affect a Collector are never destructive-by-default. Removing a Collector from a Task, or editing Task instructions after Collectors are already assigned, always confirms the action and states its effect in plain language before it takes effect.

5. Metadata integrity is trusted, not just displayed. Because BR-21/BR-22 make metadata immutable evidence, the UI never implies a system-generated field is editable, and never shows a checksum without its verified/mismatch status alongside it (Chapter 2.7, A-08).

3. Tone & Content Guidelines

All in-app copy — labels, error messages, confirmations — follows the same voice: direct, specific, and free of jargon a first-time Collector in the field wouldn't recognize.

Situation	Avoid	Use Instead
Checklist failure	"Error: precondition not met."	"Battery: 14% — charge to at least 20% before recording."
Upload failure	"Upload failed."	"Chunk 5 failed to upload. Check your connection and retry."
Permission denial	"Permission required."	"Camera access is needed to record. Enable it in Settings to continue."
Admin removing a Collector	"Are you sure?"	"Riya S. will lose access to this Task's Sessions immediately. Remove Riya?"
Session complete	"Done."	"All 6 chunks and their metadata are confirmed uploaded."

4. Interaction Patterns

4.1 Loading & Transient States

- Any wait under ~1 second shows no loading indicator at all — inserting one would be more distracting than the wait itself.
- Local Processing (C-10, the gap between Stop and the chunk being ready) always shows a determinate or indeterminate progress state, never a frozen screen, since chunking + metadata generation is a real, occasionally-longer-than-instant operation (NFR-META-01 budgets it at under 500ms, but the UI still accounts for outliers on older devices).
- Network-dependent screens (C-11, A-07) distinguish "loading current status" from "no data yet" from "you're offline" as three distinct visual states, never collapsed into one spinner.

4.2 Empty States

- A Collector with no assigned Projects sees a plain-language explanation ("No Projects assigned yet — check back once your Admin adds you to one"), never a bare empty list.
- An Admin's Projects List before their first Project is created leads directly into the "+ New Project" action, since that empty state has an obvious, single next step.

4.3 Error Recovery

- Every error state pairs a plain-language cause with a single, specific recovery action (Retry Chunk, Re-run Check, Enable in Settings) — never an error with no action attached.
- A failed upload never silently retries in a way the Collector can't see; FR-CHK/FR-SES status is always visibly "Failed" until either the retry succeeds or the Collector dismisses it (Volume 5 defines the underlying retry/backoff policy; this is the UI-visible contract on top of it).
- Destructive Admin actions (removing a Collector, deleting a Task) always show a confirmation naming the specific person or item affected, per Section 2, Principle 4.

4.4 Confirmations & Undo

- Reversible actions (reassigning a Collector, editing Task instructions) do not require a confirmation dialog — they save immediately and can be changed again just as easily.
- Irreversible or high-impact actions (removing a Collector's access, deleting a Project) always require an explicit confirmation step; there is no "undo" pattern in MVP scope, so the confirmation itself is the only safeguard.

5. Offline & Low-Connectivity UX

Since recording happens in the field (Chapter 1.1's original brief) and uploading is meant to continue while the Collector moves on to their next recording, offline handling is a first-class UX concern, not an edge case.

- Recording itself never depends on network connectivity — the checklist's network check (FR-CHK-04) only determines whether upload starts immediately or is queued, never whether recording is allowed to proceed.
- A queued chunk (pill-queued, Chapter 2.8 §6) is visually distinct from a failed one — the Collector should never mistake "waiting for a connection" for "something is broken."
- Returning to connectivity resumes queued uploads automatically and visibly (status pills update from Queued to Uploading without the Collector needing to take any action).

6. Platform Conventions (Flutter — Android & iOS)

The app is built once in Flutter for both platforms (Chapter 0.1, Vision Document); the UX guideline is to feel native enough on each platform without maintaining two divergent designs.

- Navigation gestures (swipe-back on iOS, hardware/gesture back on Android) are both honored wherever stack navigation (Chapter 2.4, Section 2) is in effect.
- The Recording Screen (C-09) intentionally overrides platform back gestures — this is the one deliberate exception, justified by BR-01–04's requirement that only Stop be reachable mid-recording, and it must be called out explicitly in Volume 6 so it isn't flagged later as a platform-convention bug.
- System permission prompts (camera, microphone, storage, location, notifications) always follow the platform's native prompt — Chapter 2.7's C-01 onboarding carousel primes the Collector beforehand but never replaces or restyles the OS-native dialog itself.

7. What This Chapter Deliberately Does Not Cover

- Pixel-level visual styling — that is Chapter 2.8 (Design System) and the wireframes in Chapter 2.6.
- Detailed accessibility requirements (contrast ratios, screen reader labels, touch target minimums) — that is Chapter 2.10, immediately following.
- Backend retry/backoff timing, export file formats, and notification delivery mechanics — those are Volume 5 (Backend Architecture) decisions; this chapter defines only the UX contract the backend must support.

VOLUME 2 — PRODUCT DESIGN

CHAPTER 2.10

Accessibility

Making the App Usable by Every Collector and Admin, Not Just the Median One

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Companion Documents	Chapter 2.8 (Design System), Chapter 2.9 (UX Guidelines)
Standard Referenced	WCAG 2.1, Level AA, adapted for native mobile

1. Purpose & Baseline

Human Archive's Collectors work in the field — often one-handed, sometimes in bright sunlight or low light, sometimes wearing gloves, and not always with perfect vision or color perception.

Accessibility here is treated as a baseline usability requirement for the actual working conditions this app is built for, not an optional pass applied at the end. This chapter's target standard is WCAG 2.1 Level AA, adapted for native mobile touch interfaces, applied consistently to both the Collector and Admin experiences.

2. Color & Contrast

2.1 Status color is never the only signal. Every status pill (queued/uploading/failed/complete, Chapter 2.8 §6) and every checklist icon (pass/fail, Chapter 2.8 §8) pairs its color with both an icon and a text label. A Collector with red-green color blindness must be able to tell "Failed" from "Complete" from the label and icon shape alone, with zero reliance on hue.

2.2 The status palette is validated, not chosen by eye. The four status colors (good #0CA30C, warning #FAB219, info/accent #2A78D6, critical #D03B3B) were checked against color-vision-deficiency separation and contrast requirements before being adopted into the Design System (Chapter 2.8, Section 2) — they are not to be substituted or tinted per-screen without re-validation.

2.3 Text contrast meets AA minimums in both themes. Body text maintains at least 4.5:1 contrast against its background, and large text (18px+ bold, or 24px+ regular — e.g. H1/H2 styles) at least 3:1, in both the light and dark themes defined in Chapter 2.8's CSS custom properties. Dark mode is a validated second pass against the dark surface color, never an automatic filter applied to the light-mode values.

2.4 Disabled states remain legible. A disabled button (btn-disabled, Chapter 2.8 §5) is visually distinguishable from an enabled one by more than opacity alone — it changes both fill and text color so the disabled state is legible rather than merely faded.

3. Touch Targets & Motor Accessibility

Element	Minimum Target	Rationale
Standard buttons, list rows, tab bar items	44 × 44 pt (iOS) / 48 × 48 dp (Android)	Platform-native minimums; already met by the Design System's default button and row padding (Chapter 2.8 §5).
Recording Screen Stop control (C-09)	≥ 64 pt/dp, generously spaced from any other element	This is the one control that must be reliably tappable one-handed, potentially mid-motion, at the end of a 10-minute recording (Chapter 1.9, BR-01–04).
Checklist retry / permission-fix actions (C-08)	Full-width button, ≥ 44pt height	A failed check under field conditions should never require a precise tap to recover from.

- No interactive element on any Collector screen requires a multi-finger gesture, a precise long-press, or a drag to operate — all primary actions are achievable with a single tap.

- The Recording Screen's chrome-free design (Chapter 2.4, Section 2) is itself a motor-accessibility decision as much as a focus decision: removing every control except Stop means there is nothing to mis-tap.

4. Screen Reader & Assistive Technology Support

- Every icon-only element (checklist pass/fail icon, status pill dot, tab bar icon) carries a text label read by the screen reader — the icon alone is never the entire accessible name (mirrors the color-not-alone rule in Section 2.1, extended to non-visual access).
- The Recording Screen's Stop control announces its state changes ("Recording — tap to stop" while active) so a screen-reader user always knows recording is in progress even without seeing the pulsing visual indicator.
- Form fields (Design System §7) associate every label with its input programmatically, and every error state (field.error) is announced when it appears — not just shown visually — so a Collector using VoiceOver/TalkBack hears "Collector Email, error: enter a valid email address" rather than silence.
- Progress indicators (upload progress, checklist re-run) expose their state as a value a screen reader can read (e.g. "62 percent"), not purely as an animated visual.

5. Focus Order & Keyboard/Switch Navigation

- Focus order on every screen follows visual top-to-bottom, left-to-right reading order — no manually reordered focus traps, except the deliberate exception below.
- The Recording Screen (C-09) intentionally limits focus to the single Stop control while recording is active; this is the same deliberate exception already flagged in Chapter 2.9, Section 6, and is treated as a focus-accessibility decision, not an oversight, so it should never be "fixed" by a later engineer unaware of BR-01–04.
- Modal screens (Checklist, Create/Edit forms, Metadata Detail) trap focus within the modal while open, and return focus to the triggering element on dismissal, consistent with standard mobile modal accessibility behavior.

6. Text Scaling & Layout Resilience

- All type in the Design System's scale (Chapter 2.8, Section 3) is specified in scalable units, not fixed pixels, so it respects the OS-level text size setting rather than overriding it.
- Layouts (checklist rows, metadata label/value rows, status pill lists) reflow vertically rather than truncating or overlapping when text is scaled up to at least 200%, per WCAG 1.4.4.
- No critical information is conveyed only through truncated text with no way to view the full value — long values (e.g. a full checksum in A-08) must remain accessible via tap-to-expand or a details view rather than being permanently cut off.

7. Accessibility by Role

7.1 Collector

Collector accessibility priorities skew toward field conditions: large touch targets, high-contrast status indication readable in direct sunlight, and screen-reader support for a Collector who may be visually impaired while still capable of operating the camera by feel and audio cues.

7.2 Admin

Admin accessibility priorities skew toward information-dense screens used for longer sessions at a desk or on a larger device: correct heading structure and landmark regions for screen-reader navigation across dashboards and metadata tables, and keyboard operability for any future web surface (Chapter 1.1, Section 8 open question).

8. Verification Checklist (Carried Into Volume 6 Build Acceptance)

- Automated contrast check passes for every text/background pair in both themes.
- Every status indicator manually verified to remain distinguishable with a color-vision-deficiency simulation filter applied.
- VoiceOver (iOS) and TalkBack (Android) manual pass completed on: Login (SH-02), Pre-Recording Checklist (C-07/C-08), Recording Screen (C-09), Upload/Sync Status (C-11), Assign Collectors (A-06), Metadata Detail (A-08).
- Text scaling verified at 100%, 150%, and 200% OS settings on the same screen list above, confirming no truncation or overlap.
- Touch target audit confirms every interactive element meets or exceeds the minimums in Section 3.